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Inter-patient arrhythmia classification with improved deep residual convolutional neural network

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ABSTRACT

Background and Objective: Early detection of arrhythmias has become critical due to the increased mortality from cardiovascular disease, and ECG is an effective tool for diagnosing cardiovascular disease and detecting arrhythmias. Classification based on ECG signal segments is more suitable for clinical application.
Methods: An improved deep residual convolutional neural network is proposed to classify arrhythmias automatically. Firstly, the overlapping segmentation method is used to segment the ECG signals in the MIT-BIH database into segments of 5 seconds in length to overcome the imbalance between classes, and these segments of the ECG signals are re-labeled. Then the discrete wavelet transform (DWT) is used to denoise these segments and the improved deep residual convolutional neural network is used for arrhythmia classification. In addition, the focal loss function is used to overcome the imbalanced classification difficulty between classes.
Results: The proposed method gives 94.54% sensitivity, 93.33% positive predictivity, and 80.80% specificity for normal segments. For the supraventricular ectopic segment, the proposed method gives 35.22% sensitivity, 65.88% positive predictivity, and 98.83% specificity. For the ventricular ectopic segment, the proposed method gives 88.35% sensitivity, 79.86% positive predictivity, and 94.92% specificity.
Conclusion: The results of this study indicate that the proposed improved deep residual convolutional neural network model trained by the training set obtained by using the overlapping segmentation method is comparable to a classical method and three state-of-art methods. In addition, the classification performance of the network model trained by focal loss as the loss function is further improved.

1. Introduction

An electrocardiogram (ECG) is a visual time series that records the electrical activity of the heart during each cardiac cycle. Its morphological characteristics can indicate the underlying symptoms of arrhythmia [1,2]. Due to the sudden and unpredictable nature of heart disease, it is of great significance to establish an effective remote real-time monitoring system that extends beyond the hospital to monitor patients' ECG signals in real-time and detect abnormal cardiac changes in time for the prevention and treatment of cardiovascular diseases [3,4].

In recent years, with the popularity of various classes of one-lead portable ECG devices such as chest straps and wristbands, the number of ECG data collected by these devices has grown rapidly. Therefore, it is very important to detect arrhythmia accurately, quickly, and reliably. The diagnostic results of arrhythmia detection and classification algorithms depend on the selection of ECG feature extraction and classifier [5], and these methods can be divided into two categories according to the different methods of feature extraction: one is arrhythmia classification based on artificial feature extraction, and the other is arrhythmia classification based on automatic feature extraction. In the first category of arrhythmia classification based on artificial feature extraction, Khorrmi et al. [6] proposed to use discrete wavelet transform (DWT), continuous wavelet transform (CWT), and discrete cosine transform (DCT) to extract features, and use multilayer perceptron (MLP) and support vector machine (SVM) as classifiers to classify arrhythmia. Karpagachelvi et al. [7] proposed using discrete wavelet transform (DWT), high-order statistics, and AR modeling

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to extract features, and an extra-limited learning machine (ELM) was used for arrhythmia classification. The baseline drift noise, power frequency noise, and high-frequency noise from the 2-lead ECG signals were first removed, then the specific features extracted from each lead ECG signal were put into the classifier for classification, and the weighted likelihood function was used to overcome the imbalance between classes [8]. Garcia et al. [9] used a complex network to extract features from temporal vectorcardiogram (TVCG) and used particle swarm optimization (PSO) to optimize parameters in the complex network. Finally, the extracted features were put into the SVM classifier together with RR Interval for classification. In [10], RR intervals, morphological and high order were manually extracted from ECG signals as features, and classified arrhythmias with LD classifier. Six types of features for ECG classification: RR intervals, morphological, statistical, high order statistics, wavelet transform, and wavelet packet entropy were extracted in [11]. Then a hierarchical XGBoost classifier was employed for classification. The ECG signal is decomposed using a dictionary formed by Stockwell, sine, and cosine functions. Furthermore, five features are extracted from each decomposed ECG: kurtosis, standard deviation, RR interval, energy, and permutation entropy. Then a least-square twin support vector machine is employed for classification. The parameters for this classifier were optimized using the artificial bee colony technique [12]. Lin et al. [13] extracted RR intervals and morphological features from filtered ECG signals and a linear discriminant classifier was employed to classify each heartbeat. In addition, the common artificial feature extraction methods include multi-feature fusion analysis [14,15], linear and nonlinear [16], and spectrum analysis [17]. However, the above methods have some problems, such as over-reliance on artificial feature accuracy and proportion allocation of multi-feature fusion. In addition, the first ECG coefficient transformation analysis and then feature extraction and classification will increase the complexity of the calculation process.

In recent years, with the rapid development of deep learning, neural network is not only applied to face detection, image denoising and other two-dimensional image processing, but also widely used in one-dimensional data processing. A Parametric Deep Energy Method (P-DEM) was presented in [18] for elasticity problems accounting for strain gradient effects. The approach is based on physics-informed neural networks (PINNs) for the solution of the underlying potential energy. Samaniego et al. [19] used Deep Neural Networks (DNNs) to solve Partial Differential Equations (PDEs) in engineering problems. These two methods rely purely on physics, naturally account for uncertainties, and do not require any input data. At the same time, the rapid growth of ECG signal data available for training means that neural networks with better classification performance and deeper network level can be constructed, which makes the study of arrhythmia classification involving deep learning more attention than ever before. The arrhythmia classification algorithm based on automatic feature extraction can automatically learn features in the neural network. A multichannel multiscale deep neural network was presented in [20] to achieve automatic classification of ECG signals without extracting features. An 11-layer deep CNN was proposed in [21] to achieve heart rate disorder quadruple classification. Yang et al. [22] proposed a CNN model called PDNet, which can effectively identify different types of arrhythmias. In PDNet, a convolutional block called PDblock, which consists of a pointwise convolutional layer and a depth convolutional layer, was designed and a loss function was used to improve the arrhythmia classification results, and good results were also achieved on the MIT-BIH database. Takalo-Mattila et al. [23] proposed an arrhythmia classification method based on a deep convolutional neural network. In this method, the authors use a high-pass filter, band rejection filter, and low-pass filter respectively to remove the baseline drift, electrical interference, and power-line interference noise in the ECG signals. Then windows are added to the ECG signals according to the marked R peak position in the database to extract the heartbeats. Finally, the heartbeats were input into the deep convolutional neural network for arrhythmia classification. Sellami et al. [24] designed a robust deep convolutional network to classify arrhythmia. In this method, the authors did not denoise the ECG signals in the database but directly extracted the heartbeats after detecting the R peak using the Pan-Tompkins method [25]. Different from other methods of intercepting heartbeats, the authors extract the current heartbeat, the current heartbeat and one previous heartbeat, and the current heartbeat and two neighboring heartbeats. Then three classes of heartbeats were input into the deep convolutional network for arrhythmia classification, and the result is that the second method of heartbeat extraction had the best classification effect. In addition, the authors also used the batch-weighted loss function to overcome the imbalance between classes. The common point of the above methods is that all of them need to detect the R peak in the ECG signal and extract the heartbeat according to the position of the R peak. Recently, some studies try to avoid extracting R peak or beat to realize ECG classification. Acharya et al. [21] directly divided ECG signals into segments with a length of 2 seconds and 5 seconds, and then input these two different datasets into the convolutional neural network respectively for classification of four classes of arrhythmias. The final experiment showed that the signal segment with a length of 5 seconds had the best classification effect. Li et al. [26] divided ECG signals into segments with a length of 10s and used densely connected convolutional neural network (DenseNet) to automatically extract features of different levels from the input ECG signals, and then classified them through the gated recurrent unit(GRU) block with attention. However, these papers did not deal with the imbalance between the classes, so there is still room for further improvement.

Given the above problems, an arrhythmia classification method based on the improved deep residual network is proposed in this paper. In this method, the ECG signals are firstly divided into segments with a length of 5s by using the overlapping segmentation method, which can effectively alleviate the problem of imbalance between classes. Then the discrete wavelet transform (DWT) is used to denoise these segments and the improved deep residual convolutional neural network is used for arrhythmia classification. In addition, the focal loss function is used to overcome the imbalanced classification difficulty between classes. The improved deep residual network proposed in this paper can not only deepen the depth of the network without degrading the performance of the network model but also multiple convolution kernels of different sizes in a convolution layer can be used to improve the perception of different scales of the convolution layer and increase the diversity of features.

The rest of this paper is organized as follows. The data preprocessing is described in Section 2. The approach put forward in this paper is presented in Section 3. Some experiments to evaluate the performance of this approach are presented in Section 4. In the end, the work of this paper is summed up in Section 5.

2. Data preprocessing

2.1. ECG database

Electrocardiogram (ECG) is the basis of Electrocardiogram (ECG) recognition, classification, and diagnosis. In this paper, 48 ECG recordings from the MIT-BIH arrhythmia database [27] published by MIT are used as the experimental database, of which each ECG record length is about 30min and the sampling frequency is 360Hz. Each record has 650000 sample points. Since all the patients in

Table 1
AAMI recommended classes for heartbeats

Class	Symbol	Members
Normal	N	N Normal beat L Left bundle branch block beat R Right bundle branch block beat e Atrial escape beat j Nodal (junctional) escape beat
Supraventricular Ectopic Beat	SVEB	A Atrial premature beat a Aberrated atrial premature beat J Nodal (junctional) premature beat S Supraventricular premature or ectopic beat (atrial or nodal)
Ventricular Ectopic Beat	VEB	V Premature ventricular contraction E Ventricular escape beat
Fusion beat	F	F Fusion of ventricular and normal beat
Unknown beat	Q	/ Paced beat f Fusion of paced and normal beat Q Unclassifiable beat

Table 2
Dataset division in this paper

Database		Patient numbers in MIT-BIH Arrhythmia database
DS1	DS11(training)	101,106,108,109,114,115,116, 119,122,209,223
	DS12(validation)	112,118,124,201,203,205,207, 208,215,220,230
DS2(testing)		100,103,105,111,113,117,121, 123,200,202,210,212,213,214, 219,221,222,228,231,232,233, 234

the four records (No.102, 104, 107, 217) were wearing cardiac pacemakers, which might cause interference to the experimental analysis, they were not considered. To be more scientific and convenient for comparison, as shown in Table 1, this paper classifies arrhythmias into 15 classes in strict accordance with the AAMI standard and further classifies them into 5 super classes, which are normal (N), Supraventricular Ectopic beat (SVEB), Ventricular Ectopic beat (VEB), Fusion beat (F) and Unknown beat (Q) [28].

To make a fair comparison, the ECG database is divided into DS1 and DS2, DS1 is further divided into DS11 and DS12 as described by deChazal et al. [8], trained with DS11, validated with DS12 in the inter-patient assessment paradigm, and experimentally tested with DS2. Table 2 shows the name list of MIT-BIH records in DS1 and DS2, respectively. DS1 and DS2 each contain 22 records out of 44 records available in the MIT-BIH database.

2.2. Segmentation and denoising of ECG signals

After dividing the database, each record should be segmented. Each segment has a duration of 5s and a length of 1800 sampling points. Then each segment of ECG signal is re-labeled and the labeling rules are as follows: if all heartbeats in a segment are normal, the segment class is normal; if there are both normal and abnormal heartbeats in a segment, the segment class is abnormal; if there are multiple classes of abnormal heartbeats in a segment, the largest number of abnormal heartbeats class is the segment class; if there are multiple classes and the same number of abnormal heartbeats in a segment, the segment class is the class of heartbeat that appears first. The number of segments of various classes obtained ordinary segmentation method is shown in Table 3.

As shown in Table 3, the amount of data in the database obtained by the traditional segmentation method is seriously imbalanced, especially the imbalance of the training set, which will lead to ineffective learning of the subsequent network and even non-convergence, which will seriously affect the final classification result. Therefore, a new segmentation strategy is adopted in this paper to alleviate the impact of the imbalance between classes by overlapping the segments, that is, the segment of a class with a smaller number of segments should overlap more with its adjacent segments. Based on the class with the largest number of segments in the case of non-overlapping segmentation, the overlap length of segments of other classes can be estimated by the following equation:

Table 3
The number of segments of various classes obtained by the ordinary segmentation method

Segment class		N	SVEB	VEB	F	Q
DS1	DS11(training)	3023	188	748	10	2
	DS12(validation)	2884	156	885	44	2
DS2(testing)		5874	477	1468	118	5

$$len = rd(L \times (1 - \frac{n}{N})) \quad (1)$$

where *len* represents the length to be overlapped between adjacent segments, *rd* represents rounding down, *L* represents the length of each segment, *n* represents the number of segments of the current class, and *N* represents the number of segments of the class with the largest number of segments. Taking class Q as an example, the number of segments of the current class Q is 4, and the number of segments of class N is the largest, which is 5904. The length of each segment is 1800 sampling points. According to Eq. (1), the overlap length *len* can be calculated as 1798 sampling points. Therefore, as shown in Fig. 1, there is a difference of 2 sampling points between the window segmenting the first segment and the window segmenting the second segment, and the overlap length between them is 1798 sampling points. The window that segments the 600th segment and the window that segments the 900th segment is the window that segments the first segment moved 1200 and 1800 sample points to the right, respectively.

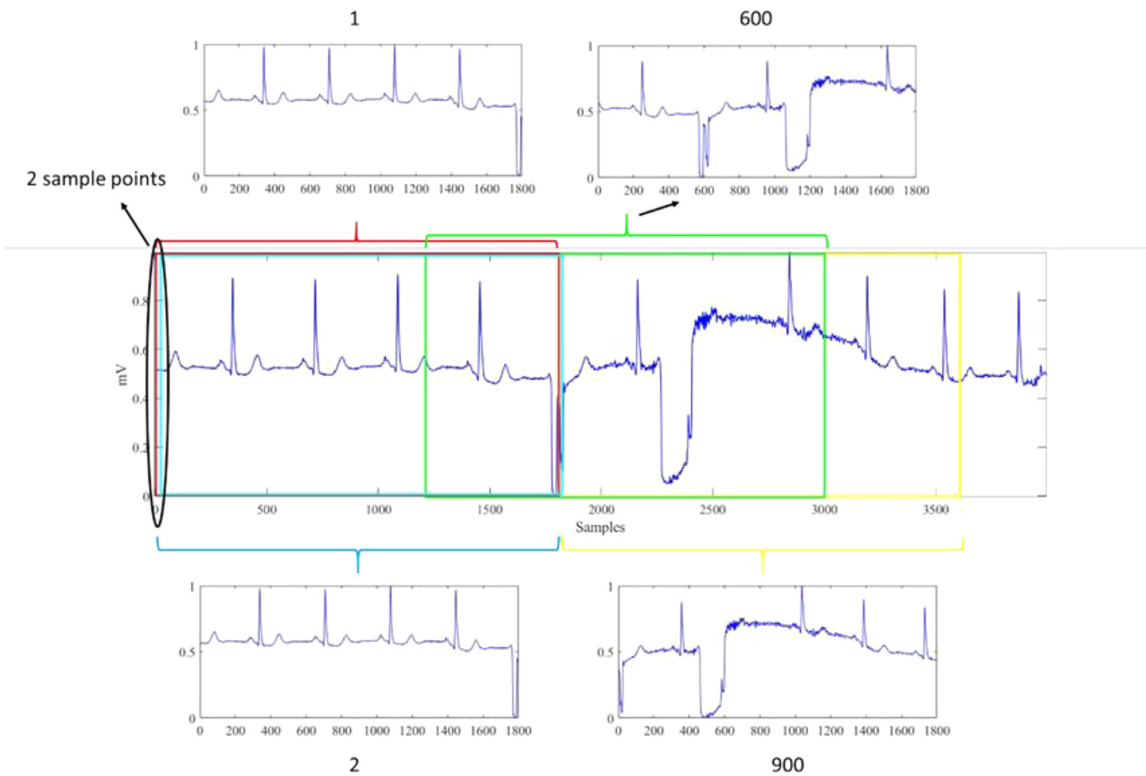

Fig. 1. Schematic diagram of overlapping segmentation of class Q

Table 4
The number of segments of various classes obtained by the overlapping segmentation method

Segment class	N	SVEB	VEB	F	Q
DS1DS11(training)	3023	3006	3010	2880	1800
DS12(validation)	2884	2843	2874	3180	1503
DS2(testing)	5874	477	1468	118	5

The number of segments of various classes obtained by using this segmentation method is shown in Table 4, where DS2 does not do overlapping segmentation as a test set. It can be seen from Table 4 that the balance between classes in the DS1 dataset after overlapping segmentation has been greatly improved compared with that in Table 3, which will be conducive to the subsequent training of the neural network model.

After segmenting the ECG signals, the discrete wavelet transform (DWT) is used to denoise each segment, and Z-score standardization is performed. The wavelet basis function is Daubechies D6(' db6 ') [29].

3. Approach

Fig. 2 presents the overall structure of the proposed approach. The approach includes dataset making unit, signal preprocessing unit, and classification model. First, ECG signals from the MIT-BIH database are used to make the dataset. The dataset-making unit includes overlapping segmenting the ECG signals and labeling each segment. Then denoising and Z-score standardization are performed on each segment in preprocessing unit. These preprocessed segments are divided into two different datasets and fed to the classification model.

3.1. Focal loss to address imbalanced classification difficulty between classes

After the above overlapping segmentation method is used to segment ECG signals, the balance between classes has been greatly improved. However, due to the small number of SVEB, F, and Q heartbeats in the MIT-BIH database, although the number of these three classes of segments in the training set has increased, it is difficult to classify them. The number of N and VEB heartbeats is big in the MIT-BIH database, so the classification difficulty is relatively low, which leads to the imbalance of classification difficulty between the five classes of segments. Therefore, the focal loss is used in this paper to overcome this problem.

The focal loss was proposed by Tsung-Yi et al. [30] to solve the problem of imbalanced classification difficulty between data for dense object detection. In object detection, the amount of data containing the object is far less than the amount of data without the object, and the classification difficulty of the data without the object is low, the improvement effect on the model is very small. The role of focal loss is to make the model pay more attention to the samples that are difficult to classify by reducing the weight of samples that are easy to classify. Therefore, the focal loss will be used in this paper to solve the problem of low difficulty in the classification of N and VEB samples, while high difficulty in the classification of SVEB, F, and Q samples. Focal Loss is introduced in this paper by introducing the cross-entropy (CE) loss function [31] used for binary classification problems. The equation of the cross-entropy loss function is as follows:

$$CE(p,y)=\begin{cases}-\log(p) & ,y=1\\ -\log(1-p) & ,y=-1\end{cases} \tag{2}$$

In Eq. (2), $p\in[0,1]$ is the estimated probability of the model, and $y\in\{\pm 1\}$ is the correct annotation class. By defining the parameters p_t, the focal loss can be extended to multiple scenarios:

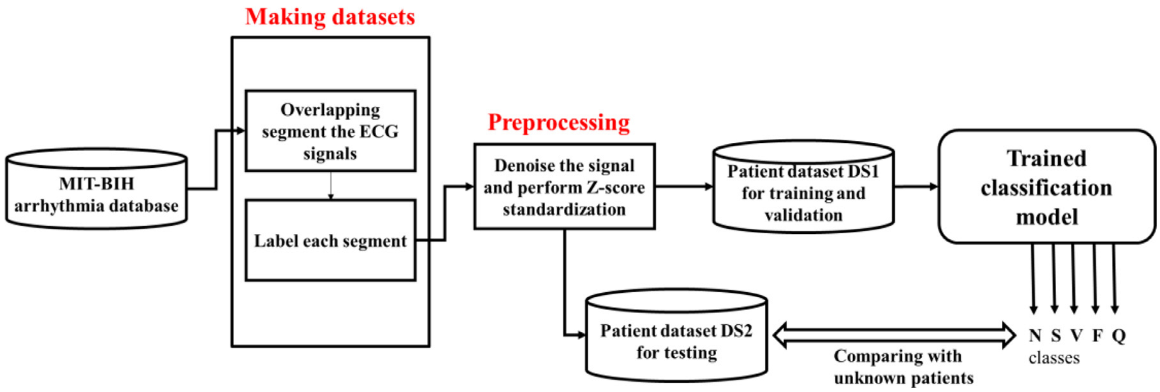

Fig. 2. Overall structure of the proposed approach

$$p_t = \begin{cases} p & ,y = 1 \\ 1 - p & ,y = -1 \end{cases} \tag{3}$$

As can be seen from Eq. (3), the CE function in Eq. (2) can be rewritten as:

$$CE(p, y) = CE(p_t) = -\log(p_t) \tag{4}$$

Eq. (5) is the CE loss function to address the class imbalance by adding a weighting factor α for class 1 and $1 - \alpha$ for class -1.

$$CE(p_t) = -\alpha_t \log(p_t) \tag{5}$$

Eq. (5) is a simple extension of the CE loss function, where α can be set by the inverse of the class frequency or set as a hyperparameter through cross-validation. Although this equation can adjust the weight of class 1 and class -1 to loss by setting the value of α, it cannot adjust the weight of samples that are easy to classify and difficult to classify. Focal loss, denoted as L_f, is an extension of the CE loss function. As shown in Eq. (6), it adds a weighted term based on the CE loss function, in which α and γ are two adjustable parameters. This equation adjusts the weight of easy and difficult classified class by adjusting the modulation coefficient γ:

$$L_f = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \tag{6}$$

where

$$\alpha_t = \begin{cases} \alpha & ,y = 1 \\ 1 - \alpha & ,y = -1 \end{cases} \tag{7}$$

where parameter γ is a fixed positive number, which is used to adjust the weight of easy to classify samples and hard to classify samples. If $\gamma = 0$, the focal loss function resembles the CE loss function. And as the value of γ increases, the efficiency of the modulation factor also increases. In addition, α is used as a fixed value between 0 and 1 to balance the positive and negative samples. The classification accuracy of the α-balanced form is better than that of the non-α-balanced form. In this paper, several experiments have provided α=0.25,$\gamma = 2$ is the optimal value.

3.2. Architecture of improved deep residual convolutional neural network

The depth of the convolutional neural network (CNN) will affect the performance of the network model. Therefore, various CNN network structures, such as AlexNet [32] and VGGNet [33], improve the performance by deepening the depth of the network structure as much as possible. However, with the increase of the depth of CNN, the model performance tends to be saturated or even degrades rapidly, which makes the training of deep networks more difficult. To solve this problem, He et al. [34] proposed the ResNet structure. ResNet makes the stacked convolution layer directly learn the residual mapping rather than the underlying mapping in the stacked network. For a stacked convolutional neural network when the input of the network is x, the feature learned by the network is $H(x)$, and another mapping-residual mapping is learned through stacked nonlinear layers:$F(x) = H(x) - x$. Then the mapping $H(x)$ learned by the original network is transformed to $F(x) + x$. Compared with direct mapping, it is easier for the network to learn the residual mapping. The specific structure of residual learning is shown in Fig. 3:

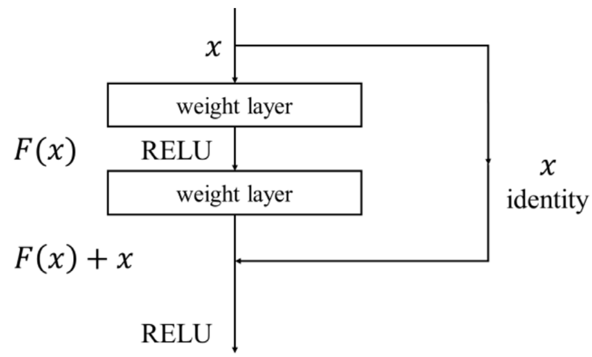

Fig. 3. The specific structure of residual learning

Formula $F(x) + x$ is realized in a feedforward network by shortcut connection, which is considered as a cross-layer connection by skipping one or more convolutional layers. In the deep residual convolutional neural network, the short-cut connection is transformed to a simple identity mapping. The identity mapping adds the input directly to the output of the stacked convolution layer. The identity mapping does not introduce additional parameters to the network, nor does it increase the computational amount and complexity of the network, and the network can still be trained by using the gradient descent method.

A convolution kernel of the same size is used in one convolution layer of the traditional CNN structure, and all features are convolved and trained on the same scale. By making multiple convolution kernels of different sizes in a convolution layer, the perception of different scales in the convolution layer can be improved and the diversity of features can be increased. Therefore, this paper proposes an improved deep residual convolutional neural network as shown in Fig. 4 and the number in parentheses in the convolution layer is the size of the convolution kernel. There are 8 residual blocks in the network structure, and each residual block has 2 convolution layers. Three kinds of convolution kernels of different sizes are adjusted in each convolution layer to adapt to different scales and the number of kernels for each size is 4k, where

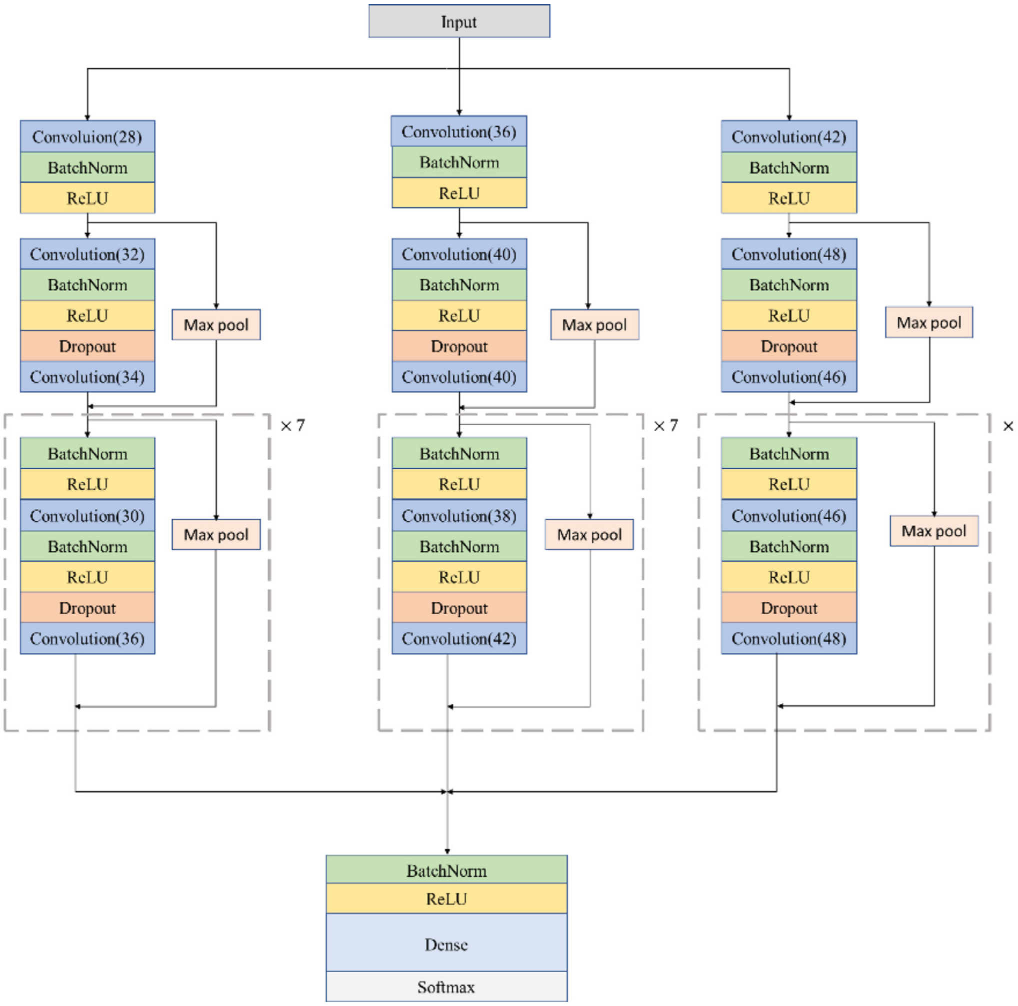

Fig. 4. Overall structure of the improved deep residual convolutional neural network (the number in parentheses in the convolution layer is the size of the convolution kernel)

k starts as 1 and is incremented every 2 convolutional layers. Every alternate residual block subsample its input by 2 times, so the original input is finally subsampled by 2^4 times. When a residual block subsamples the input, the corresponding shortcut connection also subsamples its input using the Max Pooling with the same subsampling factor. The sizes of the convolution kernel in the first convolution layer are 28, 36, and 42 respectively, and the sizes of the convolution kernel in the two convolution layers in the first residual block are 32, 40, 48 and 34, 40, 46, respectively. The convolution kernel sizes of the two convolution layers in the second to eighth residual blocks are 30, 38, 46, and 36, 42, 48, respectively.

Batch normalization [35] and a rectified linear activation (RELU) are used before each convolutional layer, and Dropout [36] is also used between the convolutional layers and after the nonlinear layer. Finally, the fully connected layer and SoftMax function output the classification results of 5 kinds of ECG signal segments. The optimizer used in this network structure is the SGD optimizer with default parameters, the loss function used here is the focal loss function given in Eq. (6).

A convolution kernel of the same size is used in one convolution layer of the Network structure in the method proposed by Sellami et al., and all features are convolved and trained on the same scale. Different from the network structure proposed by Sellami et al., the network structure proposed in this paper improves the perception of different scales in the convolution layer and increase the diversity of features by making multiple convolution kernels of different sizes in a convolution layer. In addition, the number of convolutional layers in the network structure proposed by Sellami et al is 9, and the loss function is batch-weighted loss, while the number of convolutional layers in the network structure proposed in this paper is 17, and the loss function is focal loss.

4. Results and Discussion

4.1. Performance metrics

To evaluate the performance of the model in this paper, four statistical performance metrics are adopted, namely accuracy (Acc), sensitivity (Sen), positive productivity (+P), and specificity (Spe). The equation definitions are shown in Eq. (8)-(11), and the confusion matrix obtained from the test is used for calculation.

$$Acc(\%) = \frac{TP + TN}{TP + FP + TN + FN} \times 100\% \tag{8}$$

$$Sen(\%) = \frac{TP}{TP + FN} \times 100\% \tag{9}$$

$$+P(\%) = \frac{TP}{TP + FP} \times 100\% \tag{10}$$

$$Spe(\%) = \frac{TN}{TN + FP} \times 100\% \tag{11}$$

Table 5
Classification performance results for cases of the proposed approach

	Acc	+P	Sen	Spe
A1	85.53%	49.36%	34.83%	90.51%
A2	86.41%	54.91%	51.14%	93.84%
A3	87.30%	46.48%	39.55%	94.48%
A4	88.99%	56.82%	52.10%	94.75%

where TP is true positive, indicating the number of correct classifications; FN is false negative, indicating the number of misclassifications; TN is true negative, indicating the number of heartbeats that do not belong to a certain category but are classified into this category. FP is a false positive, indicating a misclassified number belonging to a certain heartbeat. Accuracy refers to the proportion of correctly classified TP and TN in all samples, which is the most intuitive index to measure the classification effect. Sensitivity refers to the percentage of samples with positive tests in the total number of samples. The higher the sensitivity, the lower the rate of missed diagnosis. Specificity, also known as true negative rate, refers to the ratio of the number of samples classified as negative to the number of actual negative samples. The higher the specificity, the lower the rate of misdiagnosis. The positive productivity is the proportion of true positive samples in all the predicted positive samples, reflecting the possibility that the samples belong to this category. The higher the positive productivity, the lower the proportion of misdiagnosis.

4.2. Performance evaluation

In this paper, dataset DS1 is used to train the network, and DS2 is used to test network performance. To verify the effectiveness of overlapping segmentation and focal loss, as shown in Table 5, the classification results obtained in four cases (A1: without overlapping segmentation, without focal loss; A2: with overlapping segmentation, without focal loss; A3: without overlapping segmentation, with focal loss; A4: with overlapping segmentation, with focal loss) are compared in this paper. By comparing A1 and A2 in the table, when the training set is obtained by overlapping segmentation method, compared with that obtained by ordinary segmentation method, the classification accuracy increases from 85.53% to 86.41%, the positive productivity increases from 49.36% to 54.91%, the sensitivity increased from 34.83% to 52.14%, and the specificity increased from 90.51% to 93.84%. A comparison of A1 and A2 shows the effectiveness of the overlapping segmentation method. By comparing A1 and A3, when the loss function is focal loss instead of cross-entropy loss, the accuracy increases from 85.53% to 87.30%, the sensitivity increases from 34.83% to 39.55%, and the specificity increases from 90.51% to 94.48%. The positive productivity decreased slightly from 49.36% to 46.48%. According to A4 in Table 5, when the overlapping segmentation method is used to make the training set and focal loss is used as the loss function, the classification accuracy is 88.99%, positive productivity is 60.45%, sensitivity is 52.10%, and specificity is 95.24%. These performance metrics are all increased compared with A2 and A3.

Tables 6 and 7 present the confusion matrix of method A3 and A4, and Table 8 compares the positive productivity, sensitivity, and specificity of each type of ECG signal segment of method A3 and A4.

As can be seen from Table 8, compared with method A3, the performance indexes of class N, S and F in method A4 have been improved, especially the improvement of +P and Sen in class F is significant. However, because the number of class Q is too small, even if the method of overlapping segmentation is used to process the data set, class Q cannot be accurately classified. In most papers, Q samples could not be correctly classified. For example, The accuracy of [37] reached 94.61%, but Q could not be correctly classified. The accuracy of [38] reached 93%, but the F and Q categories could not be correctly classified.

In addition, to prove the effect of modifying the architecture to include multiple convolutional kernels of different sizes in one convolutional layer, two types of kernel size (single or multiple) are listed in table 9 for comparison. Thus, there are a total of two configurations (with overlapping segmentation, with focal loss) and the average performance of each configuration is presented in Table 9.

As can be seen from Table 9, the method of including multiple convolutional kernels of different sizes in one convolutional layer improves the classification accuracy from 86.28% to 88.99%, +P from 49.02% to 56.82%, Sen from 50.20% to 52.10, and Spe slightly decreases from 95.24% to 94.75%.

As shown in Table 10, to evaluate the proposed approach, the results are compared with a classical solution and three state-of-the-art solutions, which used the inter-patient paradigm and MIT-BIH Arrhythmia data.

According to the comparison in Table 10, the method proposed in this paper is slightly higher than the method proposed by deChazal et al [8] in terms of accuracy(88.99%-85.88%), and the two methods are close to each other in terms of other performance indicators. The method proposed by deChazal et al. constructed a classifier by extracting many domain-specific features from 2-lead ECG signals. The extracted features were used as input to the statistical classification model, and a third classifier was built to combine the output of the first two classifiers. Therefore, their method required a lot of computation and was not suitable for real-time arrhythmia classification on 1-lead ECG monitoring equipment. By contrast, the method proposed in this paper does not involve feature extraction in a specific domain and uses 1-lead ECG signals as input, which is more suitable for real-time arrhythmia classification.

In terms of accuracy, the method proposed in this paper is slightly lower than that proposed by Garcia et al. [9] (88.99%-92.4), and in terms of sensitivity of class S, the method proposed in this paper is much lower than that proposed by Garcia et al. (35.22%-61.96%). The method proposed in this paper is much better than the method proposed by Garcia et al (79.86%-50.85%) in positive productivity of class V. There is no significant difference between the two methods in other performance indicators. Although the method proposed by Garcia et al. has the highest accuracy among the five methods, it can only classify 3 classes of heartbeats, so cannot identify F and Q heartbeats.

The method proposed in this paper is slightly lower than the method proposed by Takalo-Mattila et al [23] in terms of accuracy(88.99%-89.91%), and the method proposed in this paper is only 35.22% in terms of sensitivity of class S, which is far lower than the method proposed by Takalo-Mattila et al(62.49%). However, in terms of the positive productivity of class V, the method proposed in this paper is far better than that proposed by Takalo-Mattila et al (79.86%-50.85%). There is no significant difference between the two methods in other performance indicators. Takalo-Mattila et al. classified cardiac beats by constructing a convolutional neural network with 3 convolutional layers. In the convolutional neural network, theoretically, with the increase of the number of convolutional layers and the deepening of the number of network layers, the performance of the model will be better and better. However, in practical application, the deepening of the network will lead to the problems of overfitting and gradient disappearance or gradient explosion, which makes the performance of the model tend to be saturated or even degraded. To solve this problem, residuals are added into the convolutional neural network in this paper to deepen the network and improve the model performance without saturating or degrading the model performance.

Table 6
The confusion matrix of method A3

Predicted label		N	S	V	F	Q
True label	N	5553	286	35	0	0
	S	151	1315	2	0	0
	V	394	18	65	0	0
	F	83	34	1	0	0
	Q	4	1	0	0	0

Table 7
The confusion matrix of the proposed approach (A4 in Table 5)

Predicted label		N	S	V	F	Q
True label	N	5553	238	78	4	1
	S	109	1297	8	54	0
	V	251	55	168	3	0
	F	33	34	1	50	0
	Q	4	1	0	0	0

Table 8
The comparison of performance results of method A3and A4 per class

		+P(%)		Sen(%)		Spe(%)	
		A3	A4	A3	A4	A3	A4
Class	N	89.78	93.33	94.54	94.54	69.44	80.80
	S	63.11	65.88	13.63	35.22	94.76	98.83
	V	79.50	79.86	89.58	88.35	99.49	94.92
	F	0.00	45.05	0.00	42.37	100.00	99.22
	Q	0.00	0.00	0.00	0.00	100.00	100.00

Table 9
The average performance of two different configurations

Configuration	Kernel size	ACC	+P	Sen	Spe
I	single	86.28%	49.02%	50.20%	95.24%
II	multiple	88.99%	56.82%	52.10%	94.75%

In terms of accuracy, the method proposed in this paper is slightly higher than that proposed by Sellami et al. [24] (88.99%-88.34%), and in terms of sensitivity of class S, the method proposed in this paper is far lower than that proposed by Sellami et al. (35.22%-82.04%).In terms of the positive productivity of class V, the method proposed in this paper is much better than that proposed by Sellami et al. (65.88%-30.44%). There is no significant difference between the two methods in other performance indicators. Sellami et al. constructed a residual convolutional neural network as mentioned above to classify heartbeats. There are 4 residual blocks in this model, and 2 convolution layers in each residual block. With the first convolution layer at the beginning of the model, there are 9 convolution layers in total. The improved residual convolutional neural network constructed in this paper is like it, but the difference is that multiple convolutional kernels of different sizes are used in one convolutional layer to improve the perception of different scales and increase the diversity of features. Although the method proposed in this paper has limited improvement in classification performance compared with the method proposed by Sellami et al., the innovation of the method proposed in this paper is that in practical application, it is not necessary to extract heartbeats from ECG signals and then classify heartbeats like other methods. Instead, it only needs to randomly segment the ECG signal into 5-second segments and diagnose whether there is arrhythmia in these signal segments. That is, the performance of the proposed method is comparable to the method proposed by Sellami et al. (requiring heartbeats extraction) for classification of arrhythmias without the need for heartbeats extraction.

The common feature of the four methods deChazal et al. (2004), Garcia et al. (2017), Takalo-Mattila et al. (2018), and Sellami et al. (2019) in Table 8 is that they all need to locate the position of R peak before classification and extract heartbeat based on the position of R peak. In Takalo-Mattila et al. (2018), the location of the heartbeat directly uses the artificial position marker in the database, which is equivalent to the addition of manual intervention in the practical application and the degree of automation is not high. In addition, deChazal et al. (2004), Garcia et al. (2017), and Sellami et al. (2019) used some R peak detection algorithms to locate R peak, which depends on the accuracy of these detection algorithms. In contrast, the method proposed in this paper only needs to use the ordinary segmentation method to segment the ECG signals into 5-second segments and input them into the trained classification model in the practical application process. In addition, the comparison between the proposed method and the other four methods in Table 8 shows that the proposed method is

Table 10
Compared methods with inter-paradigm

Work	Acc	Class(N)			Class(S)			Class(V)		
		Sen(%)	+P(%)	Spe(%)	Sen(%)	+P(%)	Spe(%)	Sen(%)	+P(%)	Spe(%)
Proposed method (A4)	88.99	94.54	93.33	80.8	35.22	65.88	98.83	88.35	79.86	94.92
DeChazal (2004)	85.88	99.16	86.86	94	38.53	75.94	95.35	81.59	77.74	98.78
Garcia (2017)[a]	92.4	94	98	82.55	61.96	52.96	97.89	87.34	59.44	95.91
Takalo-Mattila (2018)	89.91	91.89	97	76.93	62.49	55.86	98.11	89.23	50.85	94.02
Sellami (2019)	88.34	88.52	98.8	91.3	82.04	30.44	92.8	92.05	72.13	97.54

[a] Garcia (2017) can classify only 3 classes, so cannot identify F and Q heartbeats.

comparable to the other four methods in terms of various performance indicators. Therefore, this method not only has considerable classification performance but also is more suitable for clinical application.

4.3. Discussion

The purpose of this study is to propose a novel signal-lead arrhythmia classification method that does not require the detection of R peaks and heartbeats segmentation, but still has the same or better performance than existing methods. Here, we summarize the key features of our method and discuss the impact and implications of each feature on the performance and applicability of the proposed arrhythmia classification method.

Firstly, an improved deep residual convolutional neural network is proposed in this paper for arrhythmia classification. In this network architecture, the residual block proposed in ResNet [34] is used to avoid saturation or even rapid degradation of model performance while deepening the network depth as much as possible. Moreover, to increase the perception of different scales, different sizes of convolution kernels will increase the diversity of features, and discrimination can be better reflected in different kernel sizes. This novel architecture can achieve high classification performance under the given ECG data. In this method, it is not necessary to use any R peak detection algorithm to detect the R peak and extract the heartbeat to create the dataset but simply segment the ECG signal. And the method does not require feature extraction in a specific domain but automatically extracts features, which greatly improves the applicability of the method to various fields other than the classification of arrhythmias. Different from the method proposed in this paper, many existing methods rely on the methods of extracting the heartbeat and extracting the features of the heartbeat to achieve high classification performance. But even if they extract domain-specific features or extract heartbeats from ECG signals, the proposed approach is comparable to their approach for all four AAMI performance metrics in Table 10.

Secondly, as shown in Table 3, the dataset obtained by the ordinary ECG signal segmentation method is number imbalanced between classes. The number of F and Q is far less than that of N and VEB, and the number of SVEB is also less than that of N. Therefore, the method of overlapping segmentation is used to produce new datasets. As shown in Table 4, the number of classes in the training set after ECG signal segmentation is more balanced by the overlapping segmentation method, while the test set just needs to be segmented by the ordinary segmentation method. By comparing A1 and A2 in Table 5, the model trained by using the training set in Table 4 has better performance.

Thirdly, due to the imbalance in the number of different classes of heartbeats, although the number balance of various classes of ECG signal segments is improved, the difficulty of classification among different classes of segments is imbalanced. Therefore, the focal loss is used in this paper to overcome this problem. By comparing A1 and A3, A2 and A4 in Table 5, replacing the cross-entropy loss function with focal loss function can effectively solve the problem of imbalanced classification difficulty in the classification of arrhythmias and improve the classification performance.

Overall, the novel residual convolutional neural network model proposed in this paper achieves high classification performance and this model is trained by using the focal loss function with a training set obtained by the overlapping segmentation method inputs. In addition, this method does not rely on the R peak detection algorithm and feature extraction in a specific field. In practical applications, it only needs to divide the input ECG signal into 5-second segments and input them into the trained model. This has more practical application value and is more suitable for clinical applications.

However, the method proposed in this paper also has its drawbacks. In the supervised learning stage, this method requires a large database of heartbeats annotated by clinical experts, but many datasets of abnormal patterns are difficult to obtain in the medical field. In addition, due to the small number of heartbeats of F and Q in the database, the classification accuracy of these two classes is low and the classification is difficult. In the following work, other loss functions such as the batch-weighted loss function can be tried to train the model to improve this problem.

5. Conclusion

An improved deep residual convolutional neural network is proposed in this paper for arrhythmia classification. It uses raw ECG data to perform an efficient arrhythmia classification because it does not require extensive data preprocessing, such as R-peak localization and feature extraction. The experiment in this paper also shows that the network model trained by the training set obtained by using the overlapping segmentation method has better classification performance. In addition, the classification performance of the network model trained by focal loss as the loss function is further improved. The effectiveness of the proposed method is illustrated by high classification performance under inter-patient paradigms, even though single-lead raw ECG data is used only. The advantage of this method is that it can classify arrhythmias without extracting heartbeats, and the classification accuracy is comparable to that of the methods mentioned in the introduction. In practical application, the collected ECG signals only need to be randomly divided into 5-second signal segments, and these signal segments can be classified by this method. Compared with the methods mentioned in the introduction, the process of heartbeat extraction is omitted. However, because of the small amount of class Q, although overlapping segmentation has been conducted, the classification accuracy is still 0. Moreover, for class Q, the overlap is 1798 samples which means that the window is shifted only two samples each time. Thus, the high similarity of training segments can lead to poor generalization capabilities for the classifier.

The method proposed in this paper needs to use a larger annotated heartbeat database to improve the classification performance. However, publicly available annotated heartbeat databases are sorely lacking, and annotating heartbeat classes by clinical experts is expensive and time-consuming. As future work, we hope to develop a semi-supervised heartbeat classification method that uses many unlabeled heartbeat databases for automatic feature learning so that we can use the learned features as additional inputs to the CNN model or for fine-tuning of network weights. Luo et al. [39] used synthetic minority oversampling techniques to deal with the problem of unbalanced data. The method computes the nearest neighbor by implementing the Euclidean distance between the data and creates the synthesized data by multiplying the nearest neighbor by a vector containing values between 0 and 1. Specifically, a small number of samples are analyzed and simulated, and new artificial simulated samples are added to the dataset so that the categories in the original data are no longer severely unbalanced. This approach is also worth trying to work with unbalanced data sets. In addition, other loss functions such as the batch-weighted loss function can be tried to train the model to improve the imbalance between classes and Time-fractional diffusion equation

Declaration of Competing Interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Data availability statement

The data that support the findings of this study are openly available in the MIT-BIH Arrhythmia Database (https://physionet.org/content/mitdb/1.0.0/), reference number [40] and ECG-ID Database(https://physionet.org/content/ecgiddb/1.0.0/), reference number [41].

References

[1] H.V. Huikuri, A. Castellanos, R.J. Myerburg, Sudden death due to cardiac arrhythmias, N. Engl. J. Med. 345 (20) (2001) 1473–1482.
[2] M.R. Homaeinezhad, S.A. Atyabi, E. Tavakkoli, H.N. Toosi, A. Ghaffari, R. Ebrahimpour, ECG arrhythmia recognition via a neuro-SVM–KNN hybrid classifier with virtual QRS image-based geometrical features, Expert Syst. Appl. 39 (2) (2012) 2047–2058.
[3] S.-L. Guo, L.-N. Han, H.-W. Liu, Q.-J. Si, D.-F. Kong, F.-S. Guo, The future of remote ECG monitoring systems, J. Geriatric Cardiol.: JGC 13 (6) (2016) 528.
[4] W. Yin, X. Yang, L. Zhang, E. Oki, ECG monitoring system integrated with IR-UWB radar based on CNN, IEEE Access 4 (2016) 6344–6351.
[5] Z. Zhang, J. Dong, X. Luo, K.-S. Choi, X. Wu, Heartbeat classification using disease-specific feature selection, Comput. Biol. Med. 46 (2014) 79–89.
[6] H. Khorrami, M. Moavenian, A comparative study of DWT, CWT and DCT transformations in ECG arrhythmias classification, Expert Syst. Appl. 37 (8) (2010) 5751–5757.
[7] S. Karpagachelvi, M. Arthanari, M. Sivakumar, Classification of electrocardiogram signals with support vector machines and extreme learning machine, Neural Comput. Appl. 21 (6) (2012) 1331–1339.
[8] P. De Chazal, M. O'Dwyer, R.B. Reilly, Automatic classification of heartbeats using ECG morphology and heartbeat interval features, IEEE Trans. Biomed. Eng. 51 (7) (2004) 1196–1206.
[9] G. Garcia, G. Moreira, D. Menotti, E. Luz, Inter-patient ECG heartbeat classification with temporal VCG optimized by PSO, Sci. Rep. 7 (1) (2017) 1–11.
[10] F.M. Dias, H.L. Monteiro, T.W. Cabral, R. Naji, M. Kuehni, E.J.d.S. Luz, Arrhythmia classification from single-lead ECG signals using the inter-patient paradigm, Comput. Methods Programs Biomed. 202 (2021) 105948.
[11] H. Shi, H. Wang, Y. Huang, L. Zhao, C. Qin, C. Liu, A hierarchical method based on weighted extreme gradient boosting in ECG heartbeat classification, Comput. Methods Programs Biomed. 171 (2019) 1–10.
[12] S. Raj, K.C. Ray, Automated recognition of cardiac arrhythmias using sparse decomposition over composite dictionary, Comput. Methods Programs Biomed. 165 (2018) 175–186.
[13] C.-C. Lin, C.-M. Yang, Heartbeat classification using normalized RR intervals and morphological features, Math. Probl. Eng. 2014 (2014).
[14] Z. Golrizkhatami, A. Acan, ECG classification using three-level fusion of different feature descriptors, Expert Syst. Appl. 114 (2018) 54–64.
[15] S. Raj, K.C. Ray, ECG signal analysis using DCT-based DOST and PSO optimized SVM, IEEE Trans. Instrum. Meas. 66 (3) (2017) 470–478.
[16] F.A. Elhaj, N. Salim, A.R. Harris, T.T. Swee, T. Ahmed, Arrhythmia recognition and classification using combined linear and nonlinear features of ECG signals, Comput. Methods Programs Biomed. 127 (2016) 52–63.
[17] C Jinghui, L Lili, L V Wei, et al., ECG arrhythmias classification with cyclic spectral analysis, J. Front. Comput. Technol. 11 (11) (2017) 1783–1791.
[18] V.M. Nguyen-Thanh, C. Anitescu, N. Alajlan, T. Rabczuk, X. Zhuang, Parametric deep energy approach for elasticity accounting for strain gradient effects, Comput. Meth. Appl. Mech. Eng. 386 (2021) 114096.
[19] E. Samaniego, et al., An energy approach to the solution of partial differential equations in computational mechanics via machine learning: concepts, implementation and applications, Comput. Meth. Appl. Mech. Eng. 362 (2020) 112790.
[20] C.-Y. Chen, et al., Automated ECG classification based on 1D deep learning network, Methods (2021).
[21] U.R. Acharya, H. Fujita, O.S. Lih, Y. Hagiwara, J.H. Tan, M. Adam, Automated detection of arrhythmias using different intervals of tachycardia ECG segments with convolutional neural network, Inf. Sci. 405 (2017) 81–90.
[22] F. Yang, X. Zhang, Y. Zhu, PDNet: a convolutional neural network has potential to be deployed on small intelligent devices for arrhythmia diagnosis, Comput. Model. Eng. Sci. 125 (1) (2020) 365–382.
[23] J. Takalo-Mattila, J. Kiljander, J.-P. Soininen, Inter-patient ECG classification using deep convolutional neural networks, in: 2018 21st Euromicro Conference on Digital System Design (DSD), IEEE, 2018, pp. 421–425.
[24] A. Sellami, H. Hwang, A robust deep convolutional neural network with batch-weighted loss for heartbeat classification, Expert Syst. Appl. 122 (2019) 75–84.
[25] J. Pan, W.J. Tompkins, A real-time QRS detection algorithm, IEEE Trans. Biomed. Eng. (3) (1985) 230–236.
[26] L. Guo, G. Sim, B. Matuszewski, Inter-patient ECG classification with convolutional and recurrent neural networks, Biocybernetic. Biomed. Eng. 39 (3) (2019) 868–879.
[27] G.B. Moody, R.G. Mark, The impact of the MIT-BIH arrhythmia database, IEEE Eng. Med. Biol. Mag. 20 (3) (2001) 45–50.
[28] E.J.d.S. Luz, W.R. Schwartz, G. Cámara-Chávez, D. Menotti, ECG-based heartbeat classification for arrhythmia detection: a survey, Comput. Methods Programs Biomed. 127 (2016) 144–164.
[29] R.J. Martis, U.R. Acharya, L.C. Min, ECG beat classification using PCA, LDA, ICA and discrete wavelet transform, Biomed. Signal Process. Control 8 (5) (2013) 437–448.
[30] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: Proceedings of the IEEE international conference on computer vision, 2017, pp. 2980–2988.
[31] J. Brownlee, A gentle introduction to cross-entropy for machine learning, Mach. Learn. Mastery 20 (2019).
[32] A. Krizhevsky, I. Sutskever, G.E. Hinton, Imagenet classification with deep convolutional neural networks, Adv. Neural Inf. Process. Syst. 25 (2012) 1097–1105.
[33] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, arXiv preprint (2014) arXiv:1409.1556.
[34] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 770–778.
[35] S. Ioffe, C. Szegedy, Batch normalization: Accelerating deep network training by reducing internal covariate shift, in: International conference on machine learning, PMLR, 2015, pp. 448–456.
[36] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, R. Salakhutdinov, Dropout: a simple way to prevent neural networks from overfitting, J. Mach. Learn. Res. 15 (1) (2014) 1929–1958.
[37] T. Li, M. Zhou, ECG classification using wavelet packet entropy and random forests, Entropy 18 (8) (2016) 285.
[38] S. Chen, W. Hua, Z. Li, J. Li, X. Gao, Heartbeat classification using projected and dynamic features of ECG signal, Biomed. Signal Process. Control 31 (2017) 165–173.
[39] X. Luo, L. Yang, H. Cai, R. Tang, Y. Chen, W. Li, Multi-classification of arrhythmias using a HCRNet on imbalanced ECG datasets, Comput. Methods Programs Biomed. 208 (2021) 106258.
[40] G.B. Moody, R.G. Mark, The impact of the MIT-BIH arrhythmia database, IEEE Eng. Med. Biol. Mag. 20 (3) (2002) 45–50.
[41] Y. Gahi, M. Lamrani, A. Zoglat, M. Guennoun, B. Kapralos, K. El-Khatib, Biometric identification system based on electrocardiogram data, in: New Technologies, Mobility & Security, 2008, pp. 1–5.